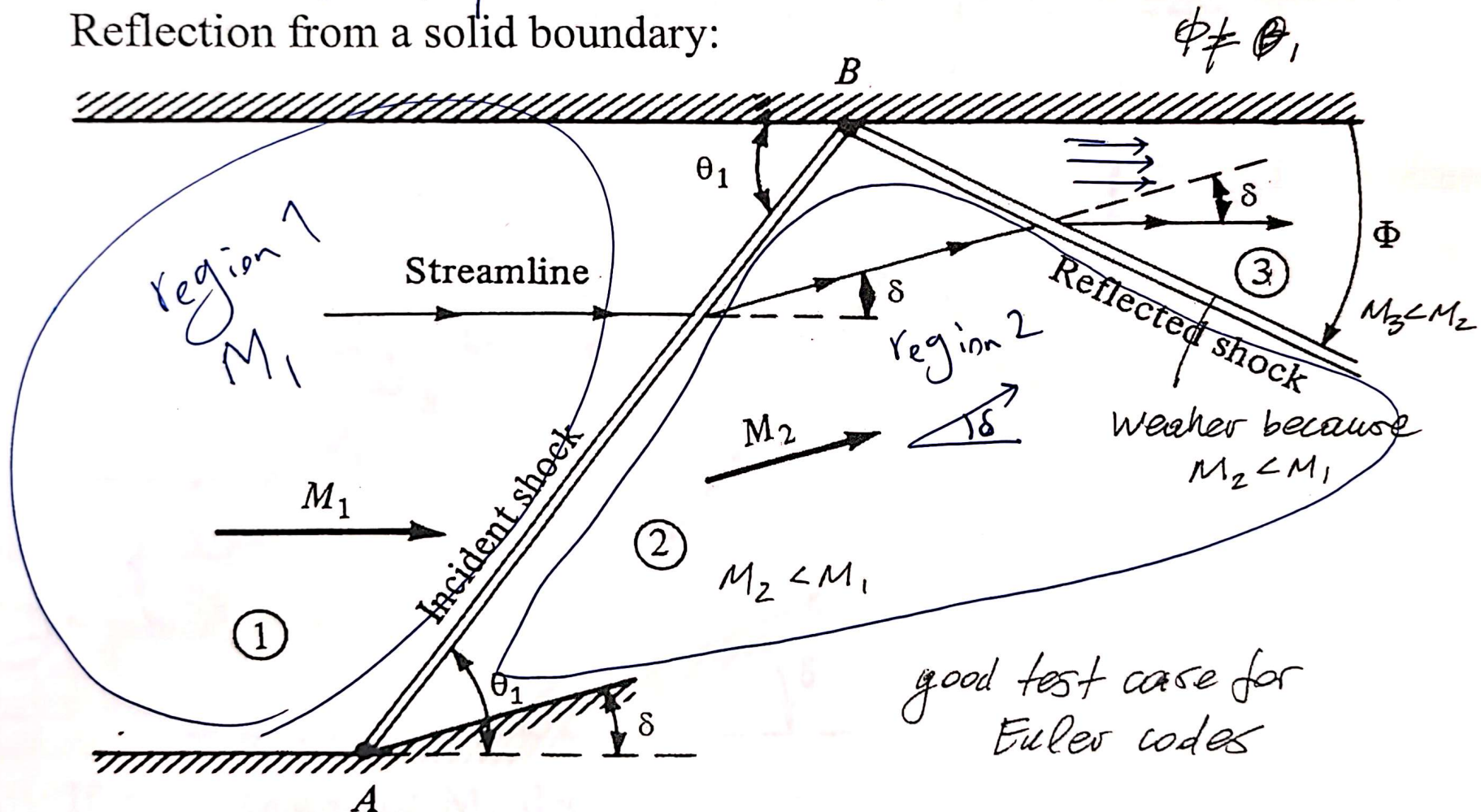


Second shock $\begin{cases} M_2 \text{ read} \\ \delta_2 = \delta_1 \end{cases} \rightarrow \theta$ First shock $\begin{cases} M_1 \rightarrow \text{read } \underline{\theta} \\ \delta_1 \end{cases}$

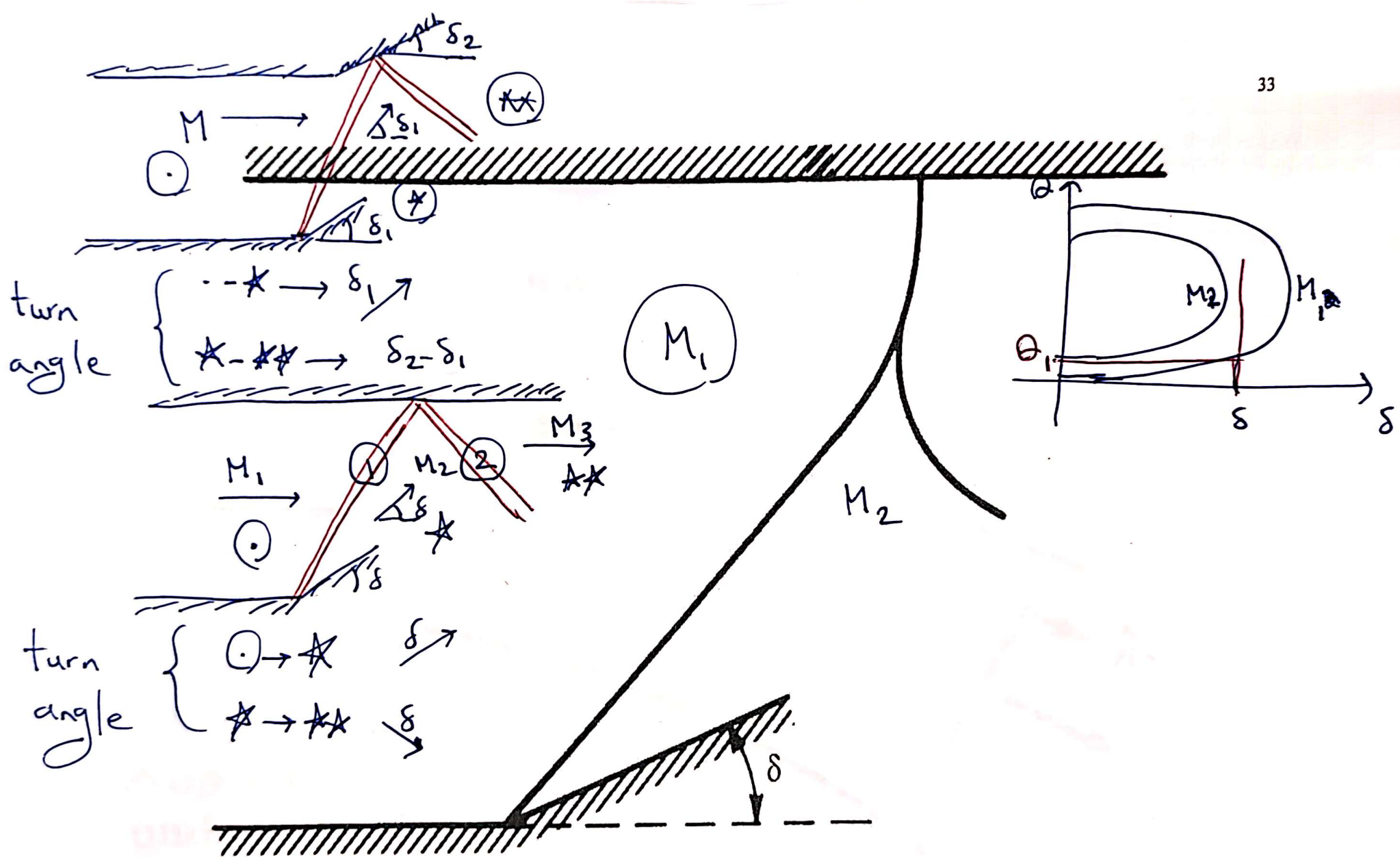
32

Reflection from a solid boundary:



2 oblique shocks sequentially; second shock is weaker $\phi \neq \theta_1$
(because it starts at a lower $M \#$)

from $M_1, \delta \rightarrow \theta_1, M_2$; $M_2, \delta \rightarrow M_3, \theta_2$ $\phi = \theta_2 - \delta$

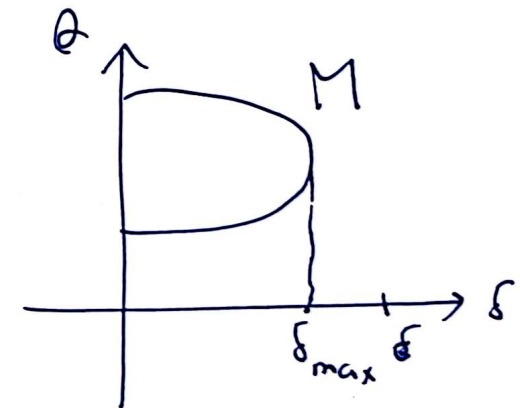
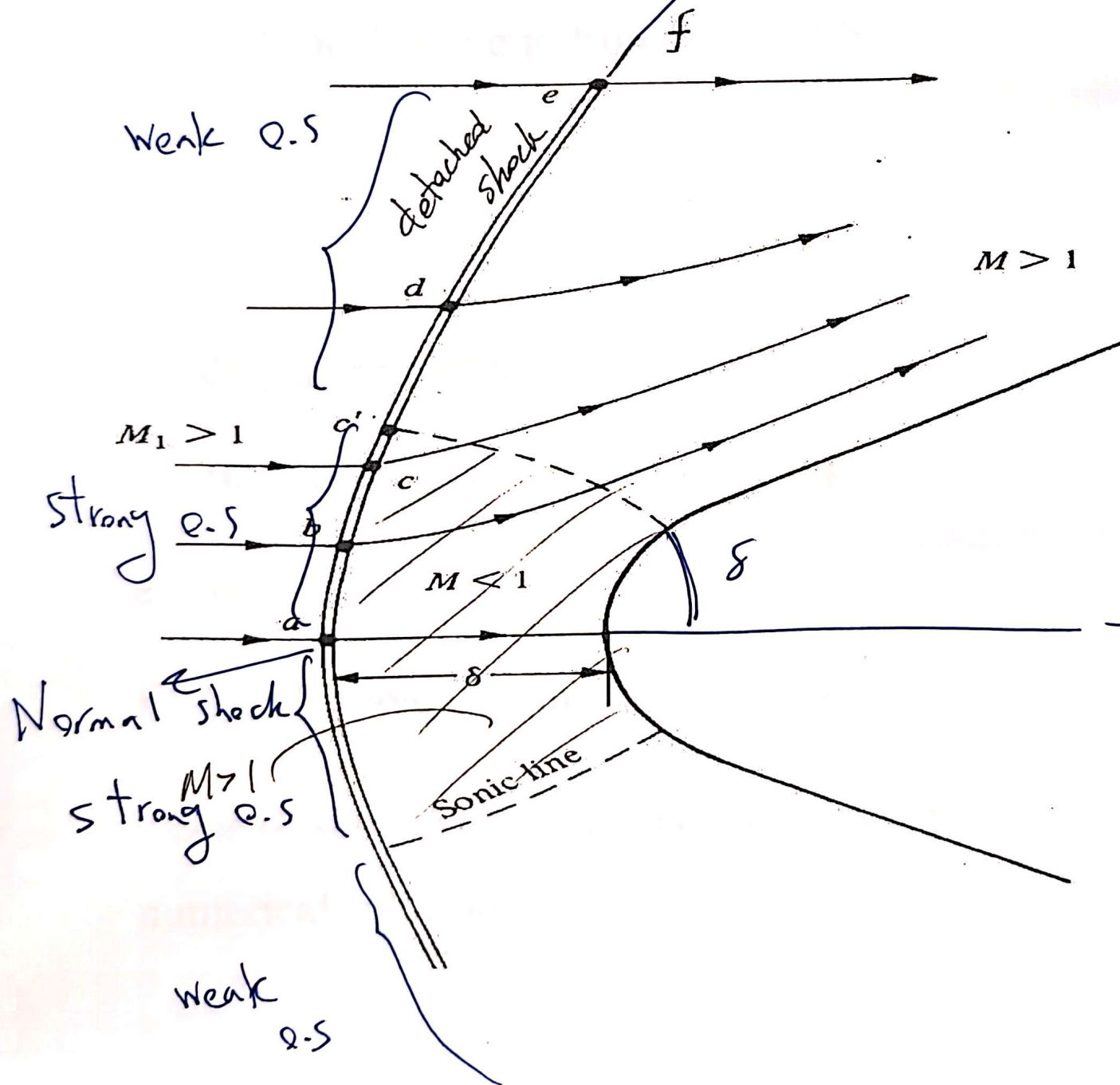


If δ increases (or M_1 decreases) a reflected shock might not be possible;

Bow shock

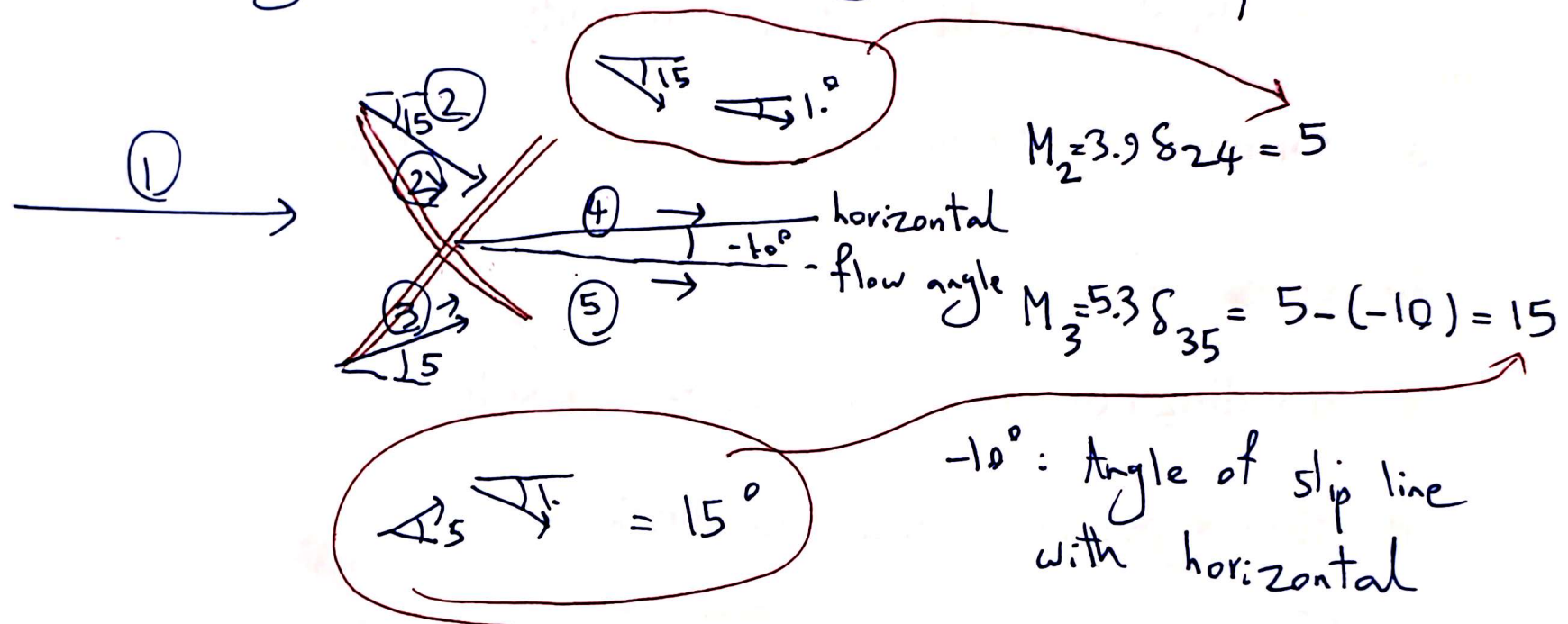
Mach wave

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$\delta > \delta_{max}$
Shock detaches

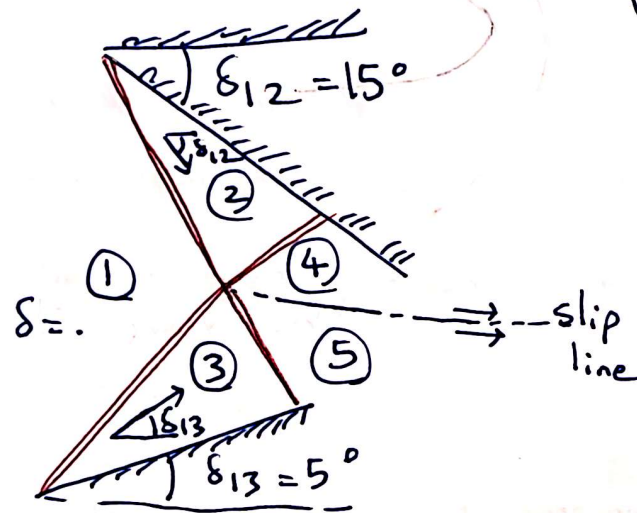
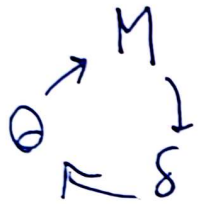
Assume an angle for flow θ in regions ④ & ⑤ θ_f



if $P_4 = P_5 \rightarrow$ our $\theta_f = -10$ was correct

θ_f	δ_{24}	$\frac{P_4}{P_2}$	$P_4 = \frac{P_4}{P_2} \frac{P_2}{P_1} P_1$	δ_{35}	$\frac{P_5}{P_3}$	$P_5 = \frac{P_5}{P_3} \frac{P_3}{P_1} P_1$
-10	5	1.619	9.832×10^{-3}	15	5.167	10.386×10^{-3}
-9.5	5.5	1.694	10.287	14.5	4.946	9.941

Do an interpolation to get $\theta P_4 = P_5 \rightarrow \theta = -9.7^\circ$



when two o.s interact,
two "are" emitted

After emitted shocks:

$$P_4 = P_5$$

the dashed line is called
slip line

slip line: entropy changes across
it but p & velocity direction
are the same same.

$$\begin{aligned} M_1 &= 6 \\ P_1 &= 10^{-3} \text{ atm} \\ \delta_{12} &= 15^\circ \\ \delta_{13} &= 5^\circ \end{aligned}$$

$$M_2 = ?$$

$$M_3 = ?$$

$$\theta_{12} = \cancel{22.6} \cancel{22.7} 22.67$$

$$\theta_{13} = \cancel{22.7} \cancel{31.6} \cancel{30.16} 13.16$$

$$M_{1n} = 6 \sin(22.67) \quad (1-2)$$

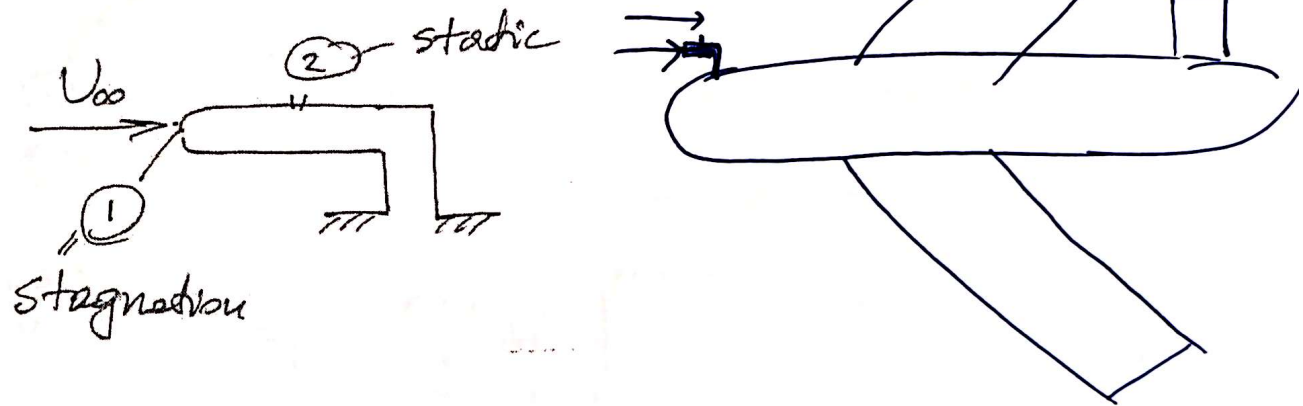
$$M_{1n} = 6 \sin(13.16) \quad (1-3)$$

$$\begin{cases} M_{1n} = 2.31 \xrightarrow{N.S} \\ M_{1n} = 1.36 \xrightarrow{N.S} \end{cases} \quad \begin{aligned} M_{2n} &= 0.533 \\ M_{3n} &= 0.757 \end{aligned} \quad \begin{aligned} M_2 &= \frac{M_{2n}}{\sin(\theta_{12} - \delta_{12})} = 3.99 \\ M_3 &= \frac{M_{3n}}{\sin(\theta_{13} - \delta_{13})} = 5.316 \end{aligned}$$

Velocity Measurement

(Pitot-static tube)

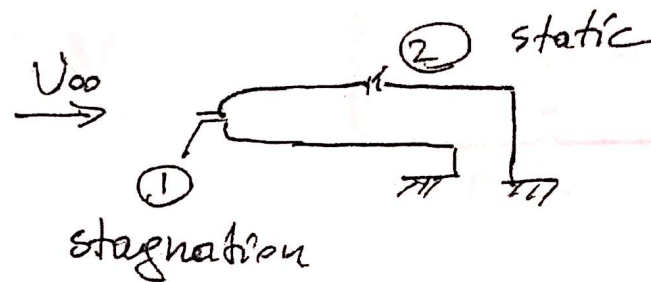
a. Incompressible (201)



Bernoulli's equation: $p_1 + \frac{1}{2} \cancel{\rho u_1^2} = p_2 + \frac{1}{2} \rho u_2^2 \rightarrow u_2 = U_{\infty}$
 $u_1 = 0$

$$\Rightarrow U_{\infty} = \sqrt{\frac{2}{\rho}(p_1 - p_2)} \quad (\text{need to know } \rho)$$

b. Compressible subsonic



Assume isentropic flow: ^(8.51)

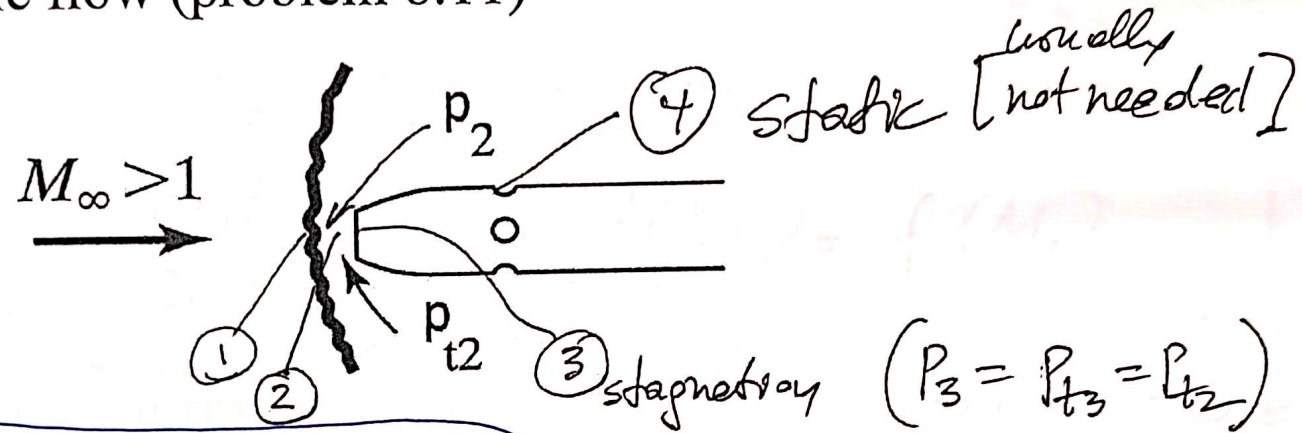
$$\frac{p_t}{p_1} = \left(1 + \frac{\gamma-1}{2} M_1^2 \right)^{\gamma/(\gamma-1)}$$

$p_t = p_2$ $M_1 = M_\infty$

$$\Rightarrow M_\infty^2 = \frac{2}{\gamma-1} \left[\left(\frac{p_1}{p_2} \right)^{(\gamma-1)/\gamma} - 1 \right]$$

Comp. Sub.

c.b: Supersonic flow (problem 8.11)



Isentropic $\frac{p_{t2}}{p_2} = \left(1 + \frac{\gamma - 1}{2} M_2^2 \right)^{\frac{\gamma}{\gamma - 1}}$

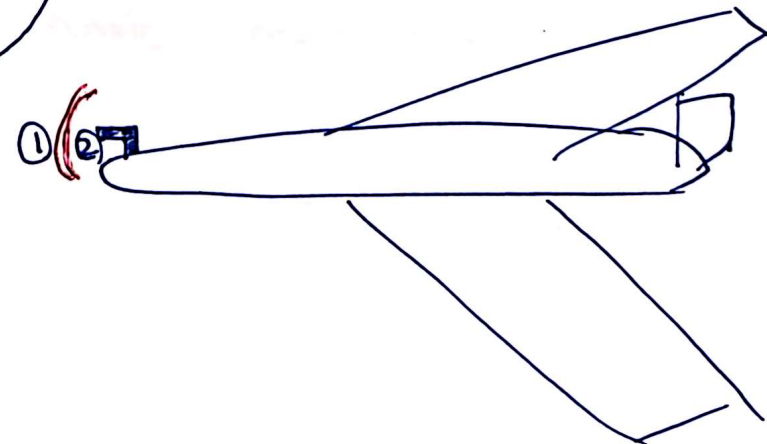
isentropic between (8.36)
 p_{t2} & p_2

Across shock normal $\frac{p_2}{p_1} = 1 + \frac{2\gamma}{\gamma + 1} (M_1^2 - 1)$

(8.75)

$M_2^2 = \frac{1 + [(\gamma - 1)/2] M_1^2}{\gamma M_1^2 - (\gamma - 1)/2}$

① ②



$$\Rightarrow \frac{p_{t2}}{p_1} = \frac{p_{t2}}{p_2} \frac{p_2}{p_1}$$

Substitute,

$$\Rightarrow \frac{p_{t2}}{p_1} = \left(\frac{(\gamma+1)^2 M_1^2}{4\gamma M_1^2 - 2(\gamma-1)} \right)^{\gamma/(\gamma-1)} \frac{1-\gamma+2\gamma M_1^2}{\gamma+1} = f(M_1) \quad (*)$$

Rayleigh-Pitot tube formula - $\frac{p_{t2}}{p_2}$ can be part of shock tables

with M_2 replaced from N.S relation based on M_1

$$p_{t2} \text{ --- } x \text{ ---}$$

$$p_2 \text{ --- } x \text{ ---}$$

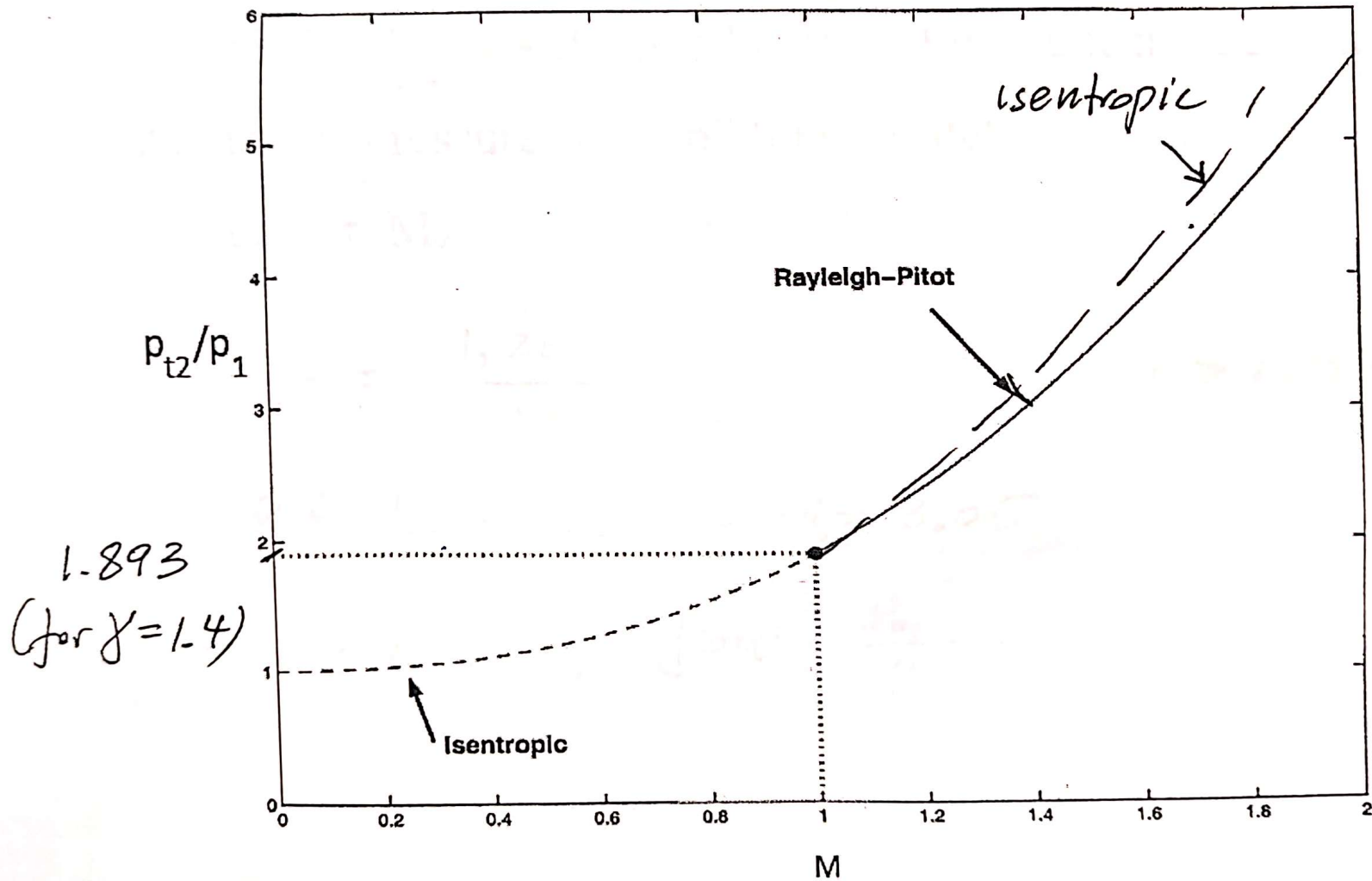
We have measured $\frac{p_{t2}}{p_2}$

and from (*) we find M_1

$$\frac{P_{t2}}{P_1} = \frac{P_{t2}}{P_{t1}} \cdot \frac{P_{t1}}{P_1} \rightarrow \left(1 + \frac{\gamma-1}{2} M_1^2\right)^{\frac{\gamma}{\gamma-1}}$$

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Rayleigh-Pitot Formula (supersonic), Isentropic Flow (subsonic)



given $\frac{P_{t2}}{P_1}$ check if $>$ or $<$ than 1.893

$\nwarrow$ R-P formula ($M > 1$) $\searrow$ isentropic ($M < 1$)

Example: A Pitot tube is placed in an air stream. The freestream pressure is $P_\infty = 1 \text{ atm} = 101,325 \text{ Pa}$. If the Pitot tube measures a value of stagnation pressure of $1,267,000 \text{ Pa}$, determine the free stream Mach number, M_∞ .

$$\frac{P_{t2}}{P_1} = \frac{1,267,000}{101,325} = 12.5043 \quad (> 1.893)$$

P-R formula: $\leadsto M_1 = 3.05$

[note: for isentropic flow = $\frac{P_{01}}{P_1} = 12.5 \leadsto M = 2.3$]
wrong answer!

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{V} A) = 0 \quad 2$$

Conservation of mass: $\rho_1 U_1 A_1 = \rho_2 U_2 A_2$

$$\nabla \cdot (\rho \mathbf{V} A) = 0$$

for 1-D flow

$$\rho U A = \text{constant} \Rightarrow d(\rho U A) = 0$$

$$\frac{\partial}{\partial x} (\rho \mathbf{V} A) = 0$$

$$\frac{(d\rho) U A}{\rho U A} + \frac{\rho (dU) A}{\rho U A} + \frac{\rho U (dA)}{\rho U A} = 0$$

$$\Rightarrow \boxed{\rho \mathbf{V} A = \text{const}}$$

$$\text{or } d(\rho \mathbf{V} A) = 0$$

$$\Rightarrow \left(\frac{d\rho}{\rho} + \frac{dU}{U} + \frac{dA}{A} \right) = 0 \rightarrow \text{Continuity Eq}^n (8.49)$$

from Momentum

Euler's equation (3.5)

$$d\left(\frac{U^2}{2}\right) + \frac{dp}{\rho} = 0$$

$$d\left(\frac{U^2}{2}\right) + \frac{dp}{\rho} = 0$$

$$\Rightarrow U dU = -\frac{dp}{\rho} = -\frac{dp}{\rho} \cdot \frac{d\rho}{d\rho} = -a^2 \frac{d\rho}{\rho}$$

Rewrite and use (8.26)

$$U dU = -\frac{dp}{\rho} = -\frac{dp}{d\rho} \cdot \frac{d\rho}{\rho} = -a^2 \frac{d\rho}{\rho}$$

$$U dU = -a^2 \frac{d\rho}{\rho}$$

$$\Rightarrow \frac{ds}{s} = -\frac{V}{a^2} dV = -\frac{V^2}{a^2} \frac{dV}{V} = -M^2 \frac{dV}{V}$$

$$\Rightarrow \frac{d\rho}{\rho} = -\frac{U dU}{a^2} = -M^2 \frac{dU}{U}$$

from continuity $\frac{ds}{s} = -\left(\frac{dV}{V} + \frac{dA}{A}\right)$

$$\left(-M^2 \frac{dV}{V} + \frac{dV}{V} + \frac{dA}{A} = 0 \Rightarrow\right)$$

put back to (8.49) $(1-M^2) \frac{dU}{U} + \frac{dA}{A} = 0$

$$\Rightarrow -\left(\frac{dU}{U} + \frac{dA}{A}\right) = -M^2 \frac{dV}{V}$$

$$\Rightarrow \frac{dA}{A} = -(1-M^2) \frac{dV}{V}$$

$$\frac{dU}{U} = -\frac{dA/A}{(1-M^2)}$$

$$\Rightarrow \frac{dV}{V} = \frac{-dA/A}{(1-M^2)} \quad (8.50)$$

1) $0 \leq M < 1$ (subsonic flow)

for $dU > 0 \Rightarrow dA < 0$

for $dU < 0 \Rightarrow dA > 0$

 converging

$V \uparrow$ as $A \downarrow$

 diverging

$V \downarrow$ as $A \uparrow$

To increase velocity you need a convergent duct

same trends as $\rho = \text{ct}$ flow

Note: for $M = 0 \Rightarrow \left(\frac{dA}{A} = -\frac{dU}{U} \right) \Rightarrow AU = \text{constant}$

$\frac{dA}{A} + \frac{dU}{U} = 0 \Rightarrow d(AU) = 0$
 $AU = \text{cte}$

2) For $M > 1$ (supersonic flow)

for $dU > 0 \Rightarrow dA > 0$

 diverging

for $dU < 0 \Rightarrow dA < 0$

 converging

To increase velocity you need a divergent duct (opposite to $M < 1$)

3) $M = 1$ (sonic flow)

$dA = 0$ for $dU \neq 0 \Rightarrow \text{max or min in } A(x)$

Actually it is a min, convergent-divergent duct